

General Relativistic Corrections to a White Dwarf Star

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21 December 2011*



Degenerate electron gas (Fermi gas)

- ★ At low temperature and high densities, the average de Broglie wavelength of a particle is much greater than the average inter-particle Separation².

$$\lambda^3 N/V \gg 1$$

Quantum effects, in particular the effects of Pauli exclusion principle become important at this limit.

- ★ This quantum effect modifies the equation of state¹

$$PV = Nk_B T \left[1 + \frac{\pi^{3/2}}{4} \frac{N h^3}{V (mk_B T)^{3/2}} \right]$$

- ★ The total wave function for fermions is anti-symmetric and it is give by the Slater determinant. Now you might notice that an additional term called exchange terms arise in the total Hamiltonian of the system. This exchange term lead to the occurrence of this additional repulsion/ pressure.

Ref: ¹K. Huang, *Statistical Mechanics*, John Wiley (1987). ² W. Greiner, L. Neise, H. Stocker, *Thermodynamics and Statistical Mechanics*, Springer (1987).

★ Low temperature system. So, how low?

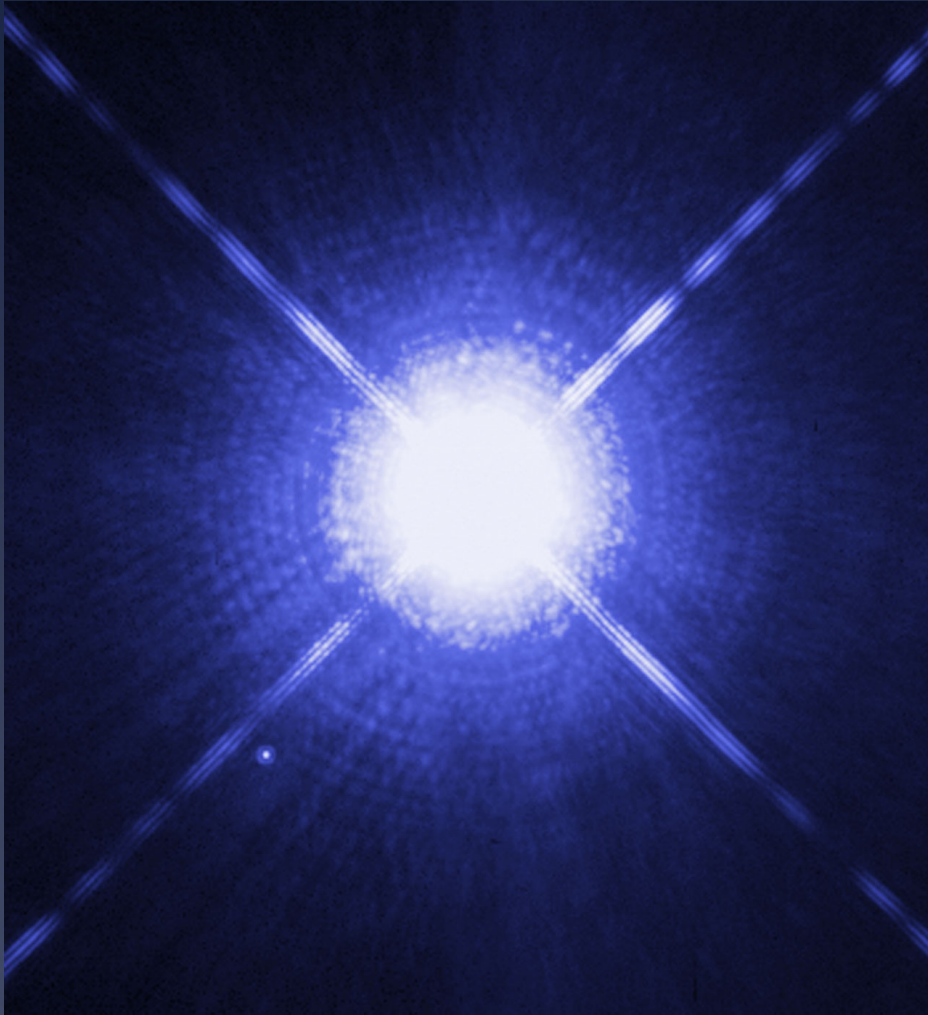
$$T \ll h^2 / mk_B (N/V)^{2/3}$$

At this temperature Fermi Dirac distribution can be approximated as step function.

$$f_{FD}(\varepsilon) = \frac{1}{e^{(\varepsilon - \mu)/kT} + 1} \approx \theta(\varepsilon) = \begin{cases} 0 & \text{if } \varepsilon > \mu \\ 1 & \text{if } \varepsilon < \mu \end{cases}$$

Such a system is said to be completely degenerate. Thus we can approximate low temperature system having high particle density to a system at absolute Zero.

What is a white dwarf star ?



- ★ It is one of the known final stages of a star. Stars below a critical value, estimated to be in the range of 6 to 8 solar masses, will finally become white dwarfs.
- ★ Such low massive star cannot proceed its nuclear reaction all the way to iron.
- ★ Temperature of the star is still high enough to keep the matter in the plasma state. The formed helium nuclei and free electron gas constituent a Typical White Dwarf star.
- ★ Temperature of the star is of the order³ $T \sim 10^{11}$ K
The Fermi temperature ($T_F \sim 10^{11}$ K) calculated from the electron density suggests that matter inside the star is highly degenerate.
- ★ The outward pressure created by the degenerate electron gas keeps the star from gravitational collapse.

Equation of state for a white dwarf star

- ★ Fowler treated the electron gas in Sirius-B as almost completely degenerate. He treated the electron gas as non-relativistic and found no limiting mass for the star.

$$E = \frac{p^2}{2m}$$

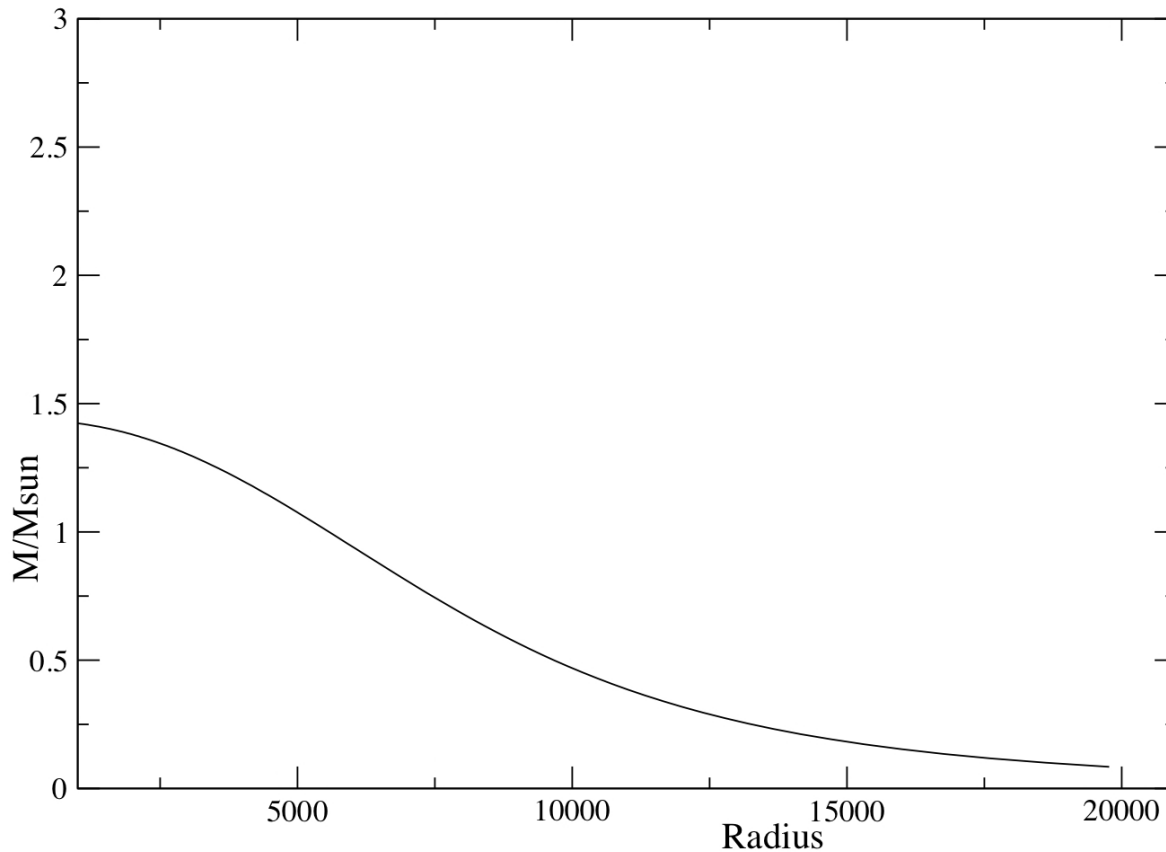
- ★ Later Anderson and Stoner considered the electron gas as relativistic and found the existence of a limiting density.

$$E^2 = p^2 c^2 + m^2 c^4$$

- ★ In 1934, Chandrasekhar succeeded in formulating the full theory of white dwarfs using the equation of state of relativistic degenerate Fermi gas⁴.

$$p = \frac{\pi m^4 c^4}{3h^3} \left[x(2x^2 - 3)(x + 1)^{1/2} + 3 \sinh^{-1} x \right]$$

$$\rho = \frac{8\pi m^3 c^3}{3h^3} x^3 \quad \text{where} \quad x = \frac{p_F}{mc}$$



Hydrostatic equilibrium equations in Newtonian limit

$$\frac{dp(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

$$\frac{dM(r)}{dr} = 4\pi\rho(r)r^2$$

- ★ Chandrasekhar use the above mentioned equation equation of state in the Hydrostatic equilibrium. He finally obtained Lane-Emden equation with index 3 Numerically solution of which suggest his famous limiting mass of $1.44 M_{\odot}$ ⁵.

Ref: ⁵S. Chandrasekhar, *Introduction to the Study of Stellar Structure*, University of Chicago Press (1939).

Einstein's theory of gravity

- ★ The action of a free particle in a curved space-time is given by⁶

$$-m \int ds \quad \text{where} \quad ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

- ★ Hamilton's principle - Particle will select a path whose action is least⁷.

$$\frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\lambda}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\lambda}{d\tau} = 0$$

- ★ Einstein's field equations⁷

$$R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R = -\frac{8\pi G}{c^4} T^{\mu\nu}$$

Tolman-Oppenheimer-Volkoff Equation

Newtonian hydrostatic equilibrium equation

$$\frac{dp(r)}{dr} = -\frac{GM(r)\rho(r)}{r^2}$$

$$\frac{dM(r)}{dr} = 4\pi\rho(r)r^2$$

They have been modified in general relativity to TOV equation

$$\frac{dp(r)}{dr} = -\frac{GM(r)}{c^2 r^2} [\varepsilon(r) + p(r)] \left[\frac{4\pi}{c^2} p(r)r^3 + M(r) \right] \left[1 - \frac{2GM(r)}{c^2 r} \right]^{-1}$$

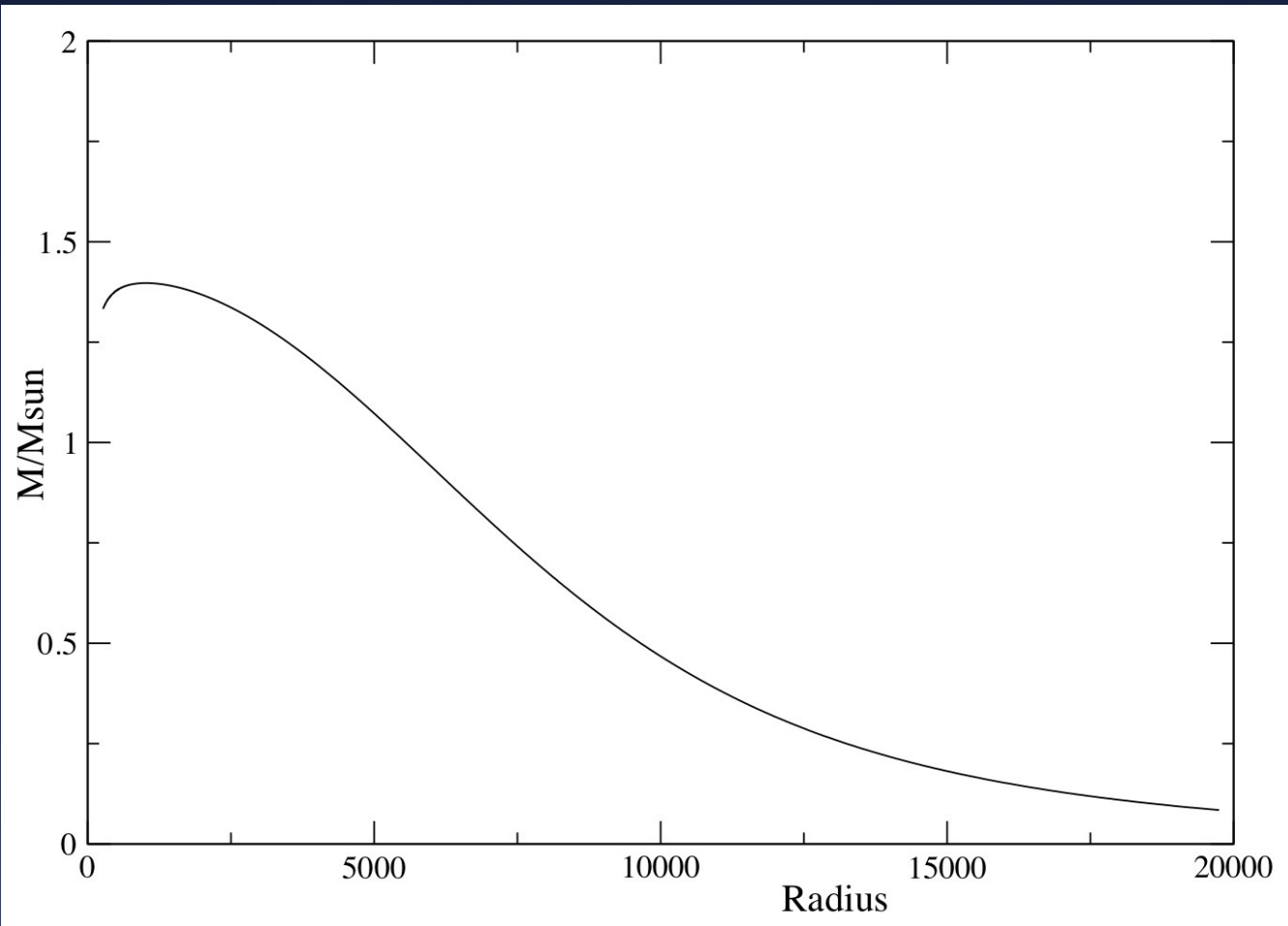
$$\frac{dM(r)}{dr} = \frac{4\pi\varepsilon(r)r^2}{c^2}$$

The new equation of state

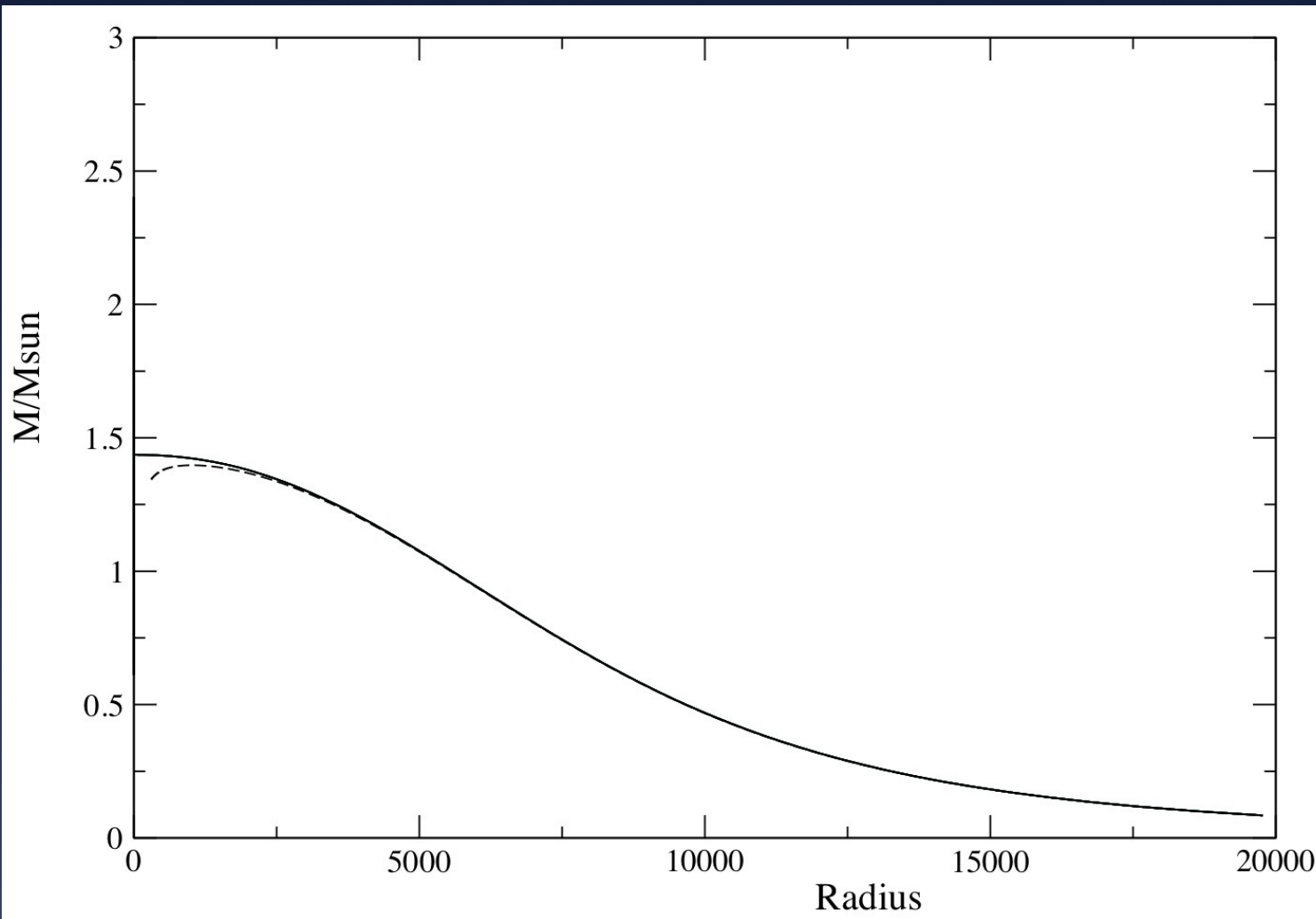
$$p = \frac{\pi m^4 c^4}{12 h^3} \left[\sinh \xi - 8 \sinh \frac{\xi}{2} + 3 \xi \right]$$

$$\rho = \frac{8 \pi m^3 c^3}{3 h^3} \sinh^3 \frac{\xi}{4} \left(1 + \frac{3 m (\sinh \xi - \xi)}{32 \mu_e H \sinh^3 \xi / 4} - \frac{m}{\mu_e} \right)$$

where $\xi = 4 \sinh^{-1} \frac{p_F}{mc}$



- ★ The star starts to collapse for higher central densities; this collapse is purely General relativistic as the equation of state contains no information about neutronization.
- ★ The plot also predicts a maximum stable mass a white dwarf can achieve, which is very nearly $1.397 M_{\odot}$.



- ★ You might notice a slight deviation from Chandrasekhar's result at higher central densities. The new value (1.397 $M_{\odot}$) that is obtained by solving the TOV equation is not too far from Chandrasekhar's limit (1.445 $M_{\odot}$).

THANK YOU